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Awareness of the alcohol–breast cancer link among breast cancer survivors in the United States: a national cross-sectional survey

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Abstract

Background The rising incidence of breast cancer in the U.S. necessitates exploration of modifiable risk factors and prevention strategies. Alcohol is a known modifiable risk factor for breast cancer; however, limited research exists on survivors' awareness and perceptions of the alcohol- breast cancer link.

Methods The Alcohol and Breast Cancer Link Awareness (ABLE) survey, conducted from September 16 to October 14, 2021, involved 5,027 female participants, including 204 breast cancer survivors. This analysis was based on a cross-sectional survey of U.S. women. Chi-square tests, Fisher's exact tests, and multinomial logistic regression were used to explore sociodemographic factors influencing awareness of the alcohol- breast cancer link among breast cancer survivors.

Results Only 30% of the breast cancer survivors were aware of the link between alcohol consumption and breast cancer. Younger women (ages 18 to 34) exhibited the highest awareness (41.7%). Notably, 26.3% of survivors were informed by a medical professional about this risk. Multivariate analyses revealed that medical advice significantly increased awareness of the alcohol-breast cancer link (Relative risk = 5.46, 95% confidence interval: 1.58–18.89, $p < 0.05$).

Conclusion Fewer than one in three breast cancer survivors are aware of the alcohol-breast cancer link, but awareness improves with medical advice. Targeted educational initiatives directed at both health care providers and older survivors are warranted.

Keywords Breast cancer survivors, Alcohol consumption, Risk awareness, Health communication, Survivorship care

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Introduction

Breast cancer mortality in the U.S. has decreased from 33.1 per 100,000 patients in 1990 to 19.1 in 2021 [1], primarily due to advancements in early detection and treatment [2–4]. However, the incidence of breast cancer increased from 126.7 per 100,000 women in 2003 to 137.4 in 2019 [1], contributing to a growing population of breast cancer survivors.

Alcohol is a Group 1 carcinogen and a modifiable risk factor linked to multiple cancer types, including breast cancer [5–10]. Pooled data from 118 studies show a slight rise in risk (1.04-fold) among light drinkers compared to nondrinkers, with moderate (1.23-fold) and heavy drinkers (1.6-fold) facing higher risks [11, 12]. Additionally, analysis of 88,000 women from two US cohorts found that light to moderate drinking was associated with a 1.13-fold increased risk of alcohol-related cancers, mostly breast cancer [13]. Research has established multiple biological pathways by which alcohol contributes to breast carcinogenesis [14]. Ethanol metabolism is a key route: the body converts alcohol into acetaldehyde, a toxic metabolite that can bind to DNA and cause mutations, initiating cancer development [15, 16]. Moreover, alcohol raises circulating estrogen levels, enhancing estrogen receptor signaling and proliferation in ER + breast cancer [17] an important factor because prolonged estrogen exposure promotes the growth of many breast tumors [15]. In addition, chronic alcohol consumption disrupts folate metabolism [18], impairing DNA repair, synthesis, and DNA methylation [5].

Importantly, alcohol consumption is common among women living with breast cancer [19–23]. A study of over 15,000 people with cancer from the All of Us Research Program found that 77.7% self-reported as past-year drinkers, and that those with alcohol-related cancers, including breast cancer, were 16% more likely to be current drinkers than people with non-alcohol-related cancers [21]. Alcohol use among survivors is relevant because alcohol may affect tumor progression and recurrence. Mechanistically, alcohol promotes oxidative stress, estrogen elevation, and immune suppression and may enhance metastasis through epithelial–mesenchymal transition (EMT) and MAPK signaling, particularly in $HER2^{+}$ breast cancers [24]. Clinically, however, the relationship between alcohol consumption and breast cancer outcomes is complex and varied [25–30]. Findings on the relationship between alcohol and recurrence risk are mixed, with some studies indicating an increased risk [30–33], especially in overweight women and those who are postmenopausal at diagnosis [34], and others showing no clear link [27, 29, 30, 35]. For example, a 2023 systematic review reported that even moderate alcohol intake may not be safe for all survivors due to increased recurrence risk [33]. Regarding mortality, a cohort study revealed that higher alcohol intake, particularly wine intake, was associated with an increased risk of death from breast cancer [36]. Furthermore, analyses of postmenopausal women diagnosed with invasive breast carcinoma and another study on breast cancer patients highlighted an increased mortality risk among those with higher alcohol consumption [37, 38]. On the contrary, some studies hint at potential benefits or no effects of post-diagnosis alcohol consumption on overall mortality and breast cancer recurrence [39–41]. Nonetheless, given robust literature on alcohol consumption increasing the risk of other cancers [42–45] and health conditions [42, 44, 45], alcohol use among breast cancer survivors may impact long-term survivorship. These findings underscore the importance of educating survivors not just about the risk of breast cancer but about alcohol's impact on overall health [21].

Recent public health surveys reveal that awareness of the alcohol–breast cancer link remains low among U.S. women, with only about 25% recognizing alcohol as a risk factor [46]. Notably, to the best of our knowledge, no studies in the U.S. have focused specifically on breast cancer survivors' awareness that alcohol use increases breast cancer risk. As breast cancer patients are engaging with medical professionals, who are often trusted sources of health information, this is an opportunity to both address the overall low level of awareness among US women about the link between alcohol use and breast cancer and to encourage healthy lifestyle behaviors among breast cancer survivors to improve their health outcomes. However, it remains unknown to what extent breast cancer survivors have ever been told by a medical professional about the link between alcohol and breast cancer, if their drinking habits changed after receiving a breast cancer diagnosis, and if these factors are associated with awareness that alcohol increases breast cancer risk. Awareness of health risks constitutes a core component in behavior change theories like the Health Belief Model [47–49], which stresses perceived threat and severity as motivational forces for avoidance behaviors [50]. Nonetheless, risk awareness may be eclipsed by emotional, habitual, or environmental elements and can fail to prevent risky conduct, as indicated by the Absent-Exempt heuristic [51] and dual-process models [52]. Thus, it is important to examine if the receipt of health information (i.e., from a medical professional) and behavioral factors (i.e., drinking behaviors post-diagnosis) are associated with awareness of alcohol as a breast cancer risk factor among breast cancer survivors to inform targeted interventions that can improve survivors' health outcomes. Given that survivors are a key group for targeted cancer treatment and support efforts, this study addresses a significant gap by assessing awareness in this population. This study aimed to (1) assess awareness among survivors that alcohol use

increases breast cancer risk, and (2) examine post-diagnosis changes in alcohol consumption and being told by a medical professional about the link between alcohol use and breast cancer as correlates of awareness among breast cancer survivors.

Methods

Participants and procedures

Data for this secondary analysis came from the cross-sectional ABLE survey, which was conducted in collaboration with Qualtrics between September 16 and October 14, 2021 [46].

All participants provided informed consent at the survey's outset. A detailed description of the study methodology has been previously described [53]. Eligibility criteria included U.S. residency, age 18 years or older, and female identification. To ensure a diverse sample with representation across age, race and ethnicity, we established quotas with Qualtrics: 20% aged 18–24, 20% 25–34, 20% 35–44, 20% 45–54, and 20% aged ≥ 55; ≥15% Black/African American; and ≥ 10% Spanish, Hispanic, or Latina. Participants received compensation at Qualtrics' discretion. The Institutional Review Board of Georgia State University approved this study (H21673).

The ABLE study included 5,027 participants. However, our focus in this secondary analysis is on the breast cancer survivor group within the study sample. Thus, our analytic sample consisted of 204 women who self-identified as having been diagnosed with breast cancer. The ABLE cross-sectional survey did not contain any questions regarding the status or stage of breast cancer or treatment.

Measures

Awareness of alcohol as a breast cancer risk factor The dependent measure, awareness that alcohol consumption increases breast cancer risk, was assessed by asking participants, "Do you think your risk of developing the following types of cancer is increased by drinking alcohol?" Participants considered eight cancer types (bladder, brain, breast, colon, liver, oral, ovarian, and stomach). This approach is consistent with previous work by Buika and colleagues examining awareness of alcohol consumption as a risk factor for breast cancer [54]. Participants responded with yes, alcohol consumption increases risk; no, alcohol consumption does not increase risk; or they were unsure or did not know, and we created a 3-level categorical variable as the outcome variable. This awareness item was adapted from a study by Buk et al. (2016) [54].

Change in alcohol use post-diagnosis To assess changes in alcohol use after receiving a cancer diagnosis, we used the query "Did your drinking change after you learned you had cancer?" and followed up with respondents who answered "yes" with the query "How did your drinking change after you learned you had cancer?", with the response options of "I increased how often I drink", "I increased how much I drink", "I reduced how often I drink", "I reduced how much I drink", "I quit drinking". We used these two items to create a 4-level categorical variable indicating no change in drinking, an increase in drinking, quitting drinking, or a reduction in drinking.

Informed by a medical professional We used the query "Has a medical professional ever told you about the link between alcohol and breast cancer?", with response options "yes", "no", or "don't remember" to create a 3-level categorical variable.

Covariates Demographic measures included age, income, education level, race and ethnicity, U.S. region of residence, and rural or urban settings. Additional covariates included mammography history, insurance status, and the Alcohol Use Disorders Identification Test (AUDIT) score. The AUDIT is a well-validated measure of alcohol use and harmful drinking patterns [55, 56]. We computed summary scores, and a created a categorical variable based on World Health Organization (WHO) cutoffs: abstainer = 0, low-risk consumption = 1–7, harmful use = ≥ 8, and possible moderate-to-severe alcohol use disorder (AUD) = ≥ 15 [55, 56].

Statistical analysis

In bivariate analysis, chi-square tests of independence and Fisher's exact test, as appropriate, were employed to identify differences in awareness by age, income, education, race and ethnicity, U.S. region, rural/urban status, AUDIT category, mammography history, insurance status, drinking change after diagnosis, and receipt of information about the alcohol-breast cancer link by a medical professional. We used multinomial logistic regression models in multivariable analysis to explore associations between awareness of the alcohol-breast cancer link and reported drinking behavior change after diagnosis, and receipt of information from a medical professional about alcohol consumption as a risk factor. In this multivariable analysis, we controlled for age, race, ethnicity, self-reported health, rural/urban status, AUDIT category and the U.S. region of residence.

Results

Among the 204 breast cancer survivors included in this analysis (out of the total survey population of 5,027 women), only 30% were aware of the alcohol- breast cancer link (Table 1). Awareness varied by age, with the highest being among younger women (41.7%, ages 18–34) ($p < 0.05$). Among respondents who reported awareness that alcohol increases breast cancer risk, 44.8% did not

Table 1 Sociodemographic characteristics, clinical characteristics, and alcohol use measures by awareness of alcohol consumption as a breast cancer risk factor among adult breast cancer survivors in the ABLE survey ($N = 204$)

	Yes, *n* (%)	No, *n* (%)	Do not know, *n* (%)	*p*-value
Outcome	60 (29.8)	74 (36.8)	67 (33.3)	
Age				
18–34	25 (41.6)	16 (21.6)	15 (22.3)	**0.022**
35–54	22 (36.6)	22 (29.7)	24 (35.8)	
55+	13 (21.6)	36 (48.6)	28 (41.7)	
Race/ Ethnicity				
Non-Hispanic White	27 (45)	43 (58.1)	38 (56.7)	0.651
Non-Hispanic Black	13 (21.6)	13 (17.5)	12 (17.9)	
Hispanic	15 (25)	15 (20.2)	10 (14.9)	
Rural/ Urban				
Rural	11 (18.3)	17 (22.9)	7 (10.4)	0.143
Urban	49 (81.6)	57 (77)	60 (89.5)	
U.S. Region				
Midwest	8 (13.3)	12 (16.2)	17 (25.3)	0.249
Northeast	15 (25)	11 (14.8)	15 (22.3)	
South	21 (35)	33 (44.5)	25 (37.3)	
West	16 (26.6)	18 (24.3)	10 (14.9)	
Insurance status				
Insured	58 (96.6)	71 (95.9)	61 (91)	0.363
Not Insured	2 (3.3)	3 (4)	6 (8.9)	
Education				
Less than high school	5 (8.4)	5 (6.8)	5 (7.6)	0.517
High school graduate	8 (13.5)	20 (27.4)	17 (26.1)	
Some college	17 (28.8)	18 (24.6)	20 (30.7)	
College or above	29 (49.1)	30 (41.1)	23 (35.3)	
Gross Income ($)				
$0 to <$24,999	19 (31.6)	29 (39.1)	23 (34.3)	0.403
$25k to <$49,999	9 (15)	19 (25.6)	14 (20.9)	
$50k+	26 (43.3)	22 (29.7)	25 (37.3)	
Drinking change after cancer diagnosis				
No change	26 (44.8)	49 (67.1)	43 (64.1)	**0.037**
Increased	20 (34.4)	13 (17.8)	9 (13.4)	
Quit	6 (10.3)	7 (9.5)	5 (7.4)	
Reduced	6 (10.3)	4 (5.4)	10 (14.9)	
Mammography history				
Yes	45 (75)	62 (83.7)	56 (83.5)	0.428
No	11 (18.3)	11 (14.8)	10 (14.9)	
AUDIT category				
Abstainer	13 (21.6)	19 (25.6)	18 (26.8)	0.181
Alcohol dependence	24 (40)	17 (22.9)	13 (19.4)	
Harmful alcohol consumption	7 (11.6)	7 (9.4)	8 (11.9)	
Low-risk alcohol consumption	16 (26.6)	31 (41.8)	28 (41.7)	
Informed by medical professional about alcohol and breast cancer link				
Yes	30 (50)	13 (17.5)	10 (14.9)	**< 0.01**
No	24 (40)	56 (75.6)	47 (70.1)	
Don't remember	6 (10)	3 (4)	10 (14.9)	

* Chi-square tests of independence and Fisher's exact test, as appropriate, were employed to calculate *p*-values for differences in awareness regarding the alcohol-breast cancer link (Yes/No/Do not know) by various outcomes

change their drinking habits post-diagnosis, compared to 34.4% who increased their intake and 10.3% who reduced/quit drinking ($p = 0.037$). Among the survivors who did not change their drinking habits post-diagnosis, 10.00% reported harmful alcohol consumption, and 16.6% were categorized as alcohol-dependent according to the AUDIT scores (Table 2). In contrast, survivors who increased their drinking post-diagnosis showed higher

Table 2 Alcohol dependence levels based on AUDIT scores in post-cancer diagnosis drinking change category among adult breast cancer survivors in the ABLE survey ($N = 204$)

Drinking change	AUDIT categories			
	Abstainer	Low-risk consumption	Harmful alcohol consumption	Alcohol dependence
Increased	7.1%	2.3%	19%	71.4%
No change	34.1%	39.1%	10%	16.6%
Quit	27.7%	72.2%	0%	0%
Reduced	9.5%	61.9%	9.5%	19%

Table 3 Multinomial model results comparing levels of awareness of alcohol consumption as a risk factor for breast cancer among cancer survivors in the ABLE survey

Awareness of alcohol as risk factor		'Yes' vs 'No'	'Yes' vs 'Do not know'	'Do not know' vs. 'No'
Characteristics	**Category**	**RR (95% CI)**	**RR (95% CI)**	**RR (95% CI)**
Age	18–25	ref	ref	ref
	35–54	1.37 (0.44, 4.23)	1.16 (0.36, 3.71)	1.18 (0.37, 3.70)
	55+	0.54 (0.15, 1.97)	0.64 (0.17, 2.43)	0.84 (0.26, 2.76)
Drinking change after diagnosis	No change	ref	ref	ref
	Increased	1.85 (0.51, 6.72)	1.13 (0.29, 4.40)	1.63 (0.42, 6.28)
	Reduced/ Quit	1.74 (0.55, 5.47)	1.08 (0.34, 3.43)	1.59 (0.55, 4.56)
Informed by medical professional about the link	No/ Don't remember	ref	ref	ref
	Yes	2.64 (0.86, 8.14)	**5.46 (1.58, 18.89)**	0.48 (0.13, 1.68)
AUDIT	Abstainer	ref	ref	ref
	Low-risk Consumption	0.56 (0.19, 2.17)	0.70 (0.22, 2.14)	0.81 (0.31, 2.06)
	Hazardous Consumption	0.35 (0.07, 1.81)	0.31 (0.05, 1.63)	1.14 (0.27, 4.82)
	Probable alcohol use disorder	0.53 (0.13, 2.17)	0.69 (0.16, 2.93)	0.77 (0.20, 2.87)

*Model adjusted for race/ethnicity, self-reported health, rural/urban status, and U.S. Region of residence

percentages of harmful alcohol consumption (19%) and alcohol dependence (71.4%). Among those who reported reducing their alcohol consumption, 9.5% reported harmful alcohol consumption and 19% reported alcohol dependence (Table 2).

Medical advice significantly influenced awareness. Of the 199 women who responded to this query, 26.6% ($n = 53$) were informed by a medical professional about the alcohol-breast cancer link (Table 1). Among these 53 women, 57% ($n = 30$) reported being aware of the link ($p < 0.01$).

Multinomial logistic regression analysis showed no statistically significant differences in sociodemographic characteristics or alcohol use measures, including post-diagnosis changes in drinking (Table 3). Being informed by a medical professional about the alcohol-breast cancer link was the only factor significantly associated with awareness of the link between alcohol use and breast cancer. Compared to survivors who were never or don't remember being informed by a medical professional about the alcohol-breast cancer link, survivors who were informed showed substantially higher odds of being aware of alcohol consumption as a risk factor for breast cancer compared to those reporting uncertainty (i.e., "don't know") about the alcohol-breast cancer link (Relative risk = 5.46, 95% confidence interval: 1.58–18.89, $p < 0.05$).

Discussion

This study assessed breast cancer survivors' awareness of alcohol as a modifiable risk factor for breast cancer and examined receipt of information about this link from a medical professional and post-diagnosis changes in drinking as correlates of awareness. To the best of our knowledge, this is the first U.S.-based study to evaluate breast cancer survivors' awareness of this link, making it a novel contribution to the literature. While public campaigns often overlook alcohol in cancer prevention and care messaging, our findings underscore the critical need to integrate this information into survivorship care. We found low awareness (30%) of the alcohol-breast cancer link among breast cancer survivors. Awareness was highest among younger women and was significantly influenced by medical advice. Despite the low overall awareness, our findings suggest that information from medical professionals may increase awareness, as observed in the higher prevalence of awareness in this sample compared to the general ABLE survey population (24%) [46]. These findings underscore the need for future studies to better understand alcohol risk perceptions among cancer survivors and what messaging and healthcare communication strategies may be most effective to increase awareness and instigate healthy behavior change. Additionally, research needs to engage health care providers to better understand their

recommendations, willingness and training in communicating health risks, including those posed by alcohol use, particularly to cancer survivors.

In our study, we assessed the relationship between changes in drinking behavior post-diagnosis and awareness of the alcohol-breast cancer link. Intriguingly, only 10% of survivors quit drinking post- diagnosis, and the remaining reported no changes (45%) or an increase in drinking (34%). A higher awareness of alcohol-breast cancer link was observed among survivors who maintained their drinking habits compared to those who increased or decreased their intake. However, these results were only significant in bivariate analyses but not in multivariate analyses. Future research should examine changes in drinking patterns in larger samples of survivors to replicate these findings and better understand the contextual issues that influence the decisions regarding alcohol use post-cancer diagnosis. While guidelines for cancer survivors mainly revolve around general cancer prevention measures, studies have revealed a link between survivors' beliefs and their proactive health changes [57, 58]. There is a compelling correlation between survivors who attribute breast cancer to alcohol consumption and their readiness to alter habits, indicating a direct connection between awareness and behavioral changes [57].

Multivariate analyses confirmed that medical professionals significantly influenced patients' awareness of the link between alcohol and breast cancer. This underscores the need for comprehensive training programs to enhance survivors' knowledge about alcohol's role in breast cancer. Several studies highlight oncologists' concerns about patients' limited awareness of this link [59–61]. These concerns often focus on medication interactions during treatment and the potential impact of alcohol on cancer recurrence post-treatment [59]. Evidence suggests survivors may not consistently receive information about alcohol-related risk from their providers. A National Cancer Institute (NCI) blog post highlighted that many survivors continue drinking, including at harmful levels, with few reporting they had been counseled on alcohol's impact on recurrence or secondary cancers [62]. Similarly, a study in the Journal of Clinical Oncology (JCO) reported that only 7% of breast cancer patients recalled receiving alcohol advice at diagnosis, and those who did were more likely to reduce intake [63]. These findings point to missed opportunities for intervention and support.

Clinicians may find it challenging to explain the concept of standard drinks and may require supportive aids. Some are hesitant to provide extensive alcohol advice, especially to patients with poor cancer prognoses, due to various barriers [59] like time constraints, competing priorities, patient disinterest, and past alcohol dependence. There are differing opinions among professionals about who should be responsible for delivering alcohol-related health messages. Integrated care points during cancer treatment have been suggested to address lifestyle factors, including alcohol intake education [59]. Interestingly, patients tend to discuss alcohol more with dietitians and breast care nurses than with oncologists, possibly perceiving them as more open to lifestyle advice discussions [60, 61]. Healthcare providers need to reinforce general cancer prevention measures and address survivors' beliefs and misconceptions by offering tailored education and guidance. This will help survivors understand specific risk factors and make informed lifestyle choices to improve cancer outcomes. The willingness of breast cancer survivors to change their alcohol intake varies by age, income, and participation in awareness events. A cohort study found that 13.2% were unwilling to reduce alcohol consumption to mitigate breast cancer risk, with higher income associated with greater reluctance to change [64]. Another study showed that receiving alcohol cessation guidance from healthcare providers significantly increased the likelihood of quitting or reducing alcohol consumption, more so than patients' own perceptions [65]. Patients who were drinking at diagnosis recalled receiving guidance from their oncologist 7% of the time (11% for excessive drinkers and 6% for those within guidelines) [65]. Perceiving the negative effects of alcohol on quality of life, fatigue, or survival was significantly associated with reduced alcohol intake post-diagnosis [65]. Counseling and information on alcohol consequences showed trends that influenced behavior change, suggesting the importance and efficacy of these interventions. Rigorous mixed-methods studies can help identify effective models for promoting behavioral changes and developing targeted interventions for alcohol moderation following cancer diagnosis [65].

Previous studies have noted that younger age at diagnosis is associated with positive post-diagnosis lifestyle changes, particularly in exercise and weight maintenance [66]. Research into cancer survivor psychology underscores the significant impact of breast cancer diagnosis on motivating healthier lifestyles [67]. However, there are gaps in seeking information among individuals without prior breast cancer symptoms, with some dismissing alcohol advice due to perceived irrelevance or lack of awareness about its association with breast cancer [68]. Breast screening and primary/urgent care clinics providing symptomatic services can serve as ideal "teachable moments" for cancer prevention advice, with positive responses to interventions [69–72]. Healthcare professionals assessing alcohol consumption among survivors found that many exceeded safe consumption guidelines, indicating the need for focused interventions [73]. Fatalistic views about cancer correlated with resistance to information on modifiable risk factors, potentially

influencing engagement in unhealthy behaviors [74]. This is relevant given that nearly all the women in our study who reported increasing their drinking post-diagnosis reported past-year drinking at levels considered harmful or dependent on alcohol. Healthcare providers play a crucial role in addressing alcohol intake by recognizing the significance of survivor demographics, psychological factors, and effective communication strategies to promote healthier lifestyles and reduce modifiable risk factors, particularly among women who drink alcohol heavily and may need additional support strategies and resources.

From a public health perspective, it is very concerning that, despite robust evidence linking alcohol consumption to an increased risk of breast cancer, public awareness of this association remains limited. A 2024 WHO report found that only 21% of European women were aware of the alcohol–breast cancer connection [15]. Similarly, a U.S. population-based survey showed only 24.4% of women recognized the risk, while over 40% were unsure. This knowledge gap is significant, especially given estimates that 4–10% of breast cancer cases are attributable to alcohol [75]. Unfortunately, little research has examined survivors' awareness, which is concerning since this population is uniquely positioned to benefit from risk-reduction education. However, we recommend that public health interventions aimed at enhancing awareness among cancer survivors to include integrating alcohol risk education into survivorship care plans and improving provider–patient communication [76]. Targeted brief advice has been shown to significantly increase the likelihood of behavior change [63]. Organizing survivorship programs to include evidence-based alcohol counseling could help reduce alcohol-related cancer recurrence [77].

The findings of this study should be interpreted with consideration of several important limitations. The cross-sectional design prevents assessment of causality or timeline verification. Although the survey was anonymous to encourage disclosure of sensitive health information, under-reporting of risky behaviors such as heavy alcohol consumption likely occurred. The small sample size of cancer survivors reduced the statistical power to detect important differences in understanding health risk behaviors, and demographic characteristics associated with the alcohol-breast cancer link awareness. Other factors and potential confounders not addressed in this study, such as access to healthcare services, psychological factors like stress or anxiety, and cultural influences on alcohol consumption and health risk awareness, could play a role. Future studies with larger sample sizes are needed to validate these findings. Ideally, prospective cohort studies following women from diagnosis onwards would offer insights into the influence of personal experiences on awareness and help tailor interventions. Our study provides new insights into an overlooked aspect of survivorship care: survivors' awareness of alcohol as a modifiable cancer risk factor. By identifying assessing the influence of medical advice, this research offers practical direction for targeted educational strategies. Future efforts should prioritize structured interventions within oncology and primary care settings to improve survivor outcomes.

Conclusion

This study assessed breast cancer survivors' awareness of the link between alcohol consumption and breast cancer, identifying age as the sole significant sociodemographic factor influencing this awareness, and suggesting older breast cancer survivors are an especially important age group in need of improved awareness. Additionally, reductions in post-diagnosis drinking were minimal, with only 10% of survivors quitting drinking, with a sizeable proportion reporting increases in intake, and no association between post-diagnosis drinking changes and awareness. This suggests the need for providers to make a clearer connection between alcohol use and breast cancer outcomes to improve awareness of alcohol's effects on breast cancer and the adoption of healthy lifestyle behaviors after a cancer diagnosis. Overall, our findings highlight the need for tailored education and intervention strategies to enhance awareness and promote healthier lifestyle choices among breast cancer survivors. It also underscores the importance of healthcare professionals in advising women in general and breast cancer survivors in particular about the risks of alcohol use.

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Author contributions
RA, VG and GS conceptualized the research project and proposed the aims for the secondary analysis; RA, PM and MS conceptualized and developed the original ABLE survey; PM and NK guided the analysis strategy; GS and VG were involved in the data management; GS performed the data analysis; GS wrote the first draft of the manuscript; PM, MS, KG, VG, and SA reviewed and edited the manuscript; NK mentored GS for statistical analysis; RA mentored GS and VG; RA provided oversight of manuscript preparation, statistical analyses and coordination of research team and revised the manuscript. All authors contributed to the interpretation of the results, reviewed the manuscript, and agreed with the final version. The work reported in the paper has been performed by the authors, unless clearly specified in the text. All authors contributed to the interpretation of the results, reviewed the manuscript, and agreed with the final version. The work reported in the paper has been performed by the authors, unless clearly specified in the text.

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Data availability
The ABLE survey dataset analyzed in this study is available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

All relevant ethical safeguards have been met in relation to patient or subject protection. The study has complied with the World Medical Association Declaration of Helsinki. The study was approved by the Georgia State University Institutional Review Board (H21673). Informed consent was obtained from all participants, at the survey's outset.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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